

30 – Frontend Foundations and UI Module Designs

The two user-interface shells (web back office and Android till), the shared design system built around “screens as paper”, the usability rules for low-literacy trilingual users, and a screen inventory for each of the nine modules so that a full-stack developer can take a module end to end.

Field	Value
Document	30 – Frontend Foundations and UI Module Designs (design artefact; companion to 20 and to each module design 21–29)
Version	v0.1 – draft for design review
Date	6 September 2026
Status	Draft. The till technology stack (Section 4) and the wireframing toolchain are decision requests. Screen inventories are the handover baseline for module developers and are refined inside each module design.
Inputs	05 SRS, 06 NFR (usability directive), 07 Matrices (roles, configuration register), 16 Questionnaire Answers (H, N, O, P, S), 18 Core Data Model, 19 Cross-Cutting Services (identity, permissions, i18n), 20 Stage 2 plan v0.2; external: Material Design 3 accessibility guidance, UnifiedPOS device categories, WCAG 2.2, retail POS usability practice.
Audience	Full-stack module developers, the platform pair owning the shells, UX reviewer, federation training lead.

The one rule that shapes every screen. Shop staff have generally low computer familiarity (S-02) and already know the paper: the requisition book, the delivery note, the GRN, the khata, the day-end sheet. Every screen mirrors a paper document they already understand, laid out the way the paper is laid out, with the same words in their language. Software vocabulary (dashboard, entity, sync, transaction) never appears on a shop screen.

1. Surfaces and who uses them

Surface	Users	Technology	Connectivity	Owner
Web back office	Federation staff, distributor and MPCs office staff, auditors, regulator	React + TypeScript single-page application; responsive down to phone width (O-06)	Online only	Platform pair (shell); modules (screens)
Till application	Cashiers, supervisors, shop-in-charge; shop-management mode for receiving, counts, write-off drafts	Android native, kiosk-locked, local database, peripheral layer (Section 4)	Offline-first, 1–2 days	M6 (shell and sales); M4/M5 (shop-mode screens)
Driver application	Delivery drivers: drop list, proof of delivery, photos	Same Android stack, minimal	Offline-capable, syncs like a till	M4
Customer display	Shoppers at the counter: line, total, change	Secondary screen of the till where present	–	M6
Printed output	Receipts, Z-reports, A4 documents and reports	Till raster renderer; server A4 pipeline	–	M6, M8

There is deliberately no separate mobile application for managers: the responsive web back office covers phones (O-06), and the till's shop-management mode covers shop tasks on the one device a small shop has.

2. Design system: screens as paper

2.1 Foundations

Aspect	Design
Typography	Noto Sans, Noto Sans Sinhala, Noto Sans Tamil bundled with both shells; one type scale (12/14/16/20/28/40 px) where 40 px is for money totals at the till. Line height generous for Sinhala and Tamil stacked glyphs (1.5).
Layout grid	8 px base. Till: three fixed zones (Section 4.3). Web: document-shaped pages, single column on phones.
Touch and click targets	Minimum 48 × 48 dp on the web (Material and WCAG 2.2 guidance); 56 × 56 dp for primary till actions and every keypad key; 12 dp minimum spacing between adjacent targets.
Colour	Neutral surfaces; one accent; semantic colours for state (draft, issued, disputed, void, alert) always paired with an icon and a label – colour is never the only signal (P-08). Contrast ratio at least 4.5:1 for text, 3:1 for large text and icons.
Iconography	One icon set; every icon carries a visible label on shop screens; icons follow the paper metaphor (book, lorry, scale, stamp), not software metaphors (gear, cloud).
Language	All strings via the message catalogue (doc 19 section 5.2); three-column names with field-level English fallback marked by a small “EN” tag so staff know the translation is missing rather than the item being foreign.
Numbers and money	Western Arabic digits everywhere (doc 19 DR-6); money as Rs 1,234.00 with tabular figures; quantities with unit suffix (12 kg, 3 × 24).
Feedback	Every completed action gives three signals at the till: a sound, a colour flash on the affected line, and a text confirmation; on the web, a toast plus the document's state chip changing.

2.2 Shared components (the paper vocabulary)

Component	What it is	Used by
ScopeBanner	Persistent strip showing the entity and location the user is acting for, and the policy class if not OWN (Federation view, external). Prevents acting in the wrong scope.	Both shells, every screen
DocumentHeader / DocumentLines	The paper form: number, date, parties, status chip; a line table with SKU name (three-column aware), batch, quantity, unit, price. Read-only after issue; drafts editable inline.	M4, M5, M6, M7
StateChip + Timeline	Current state from the state machine; a vertical timeline of transitions from document_state_history with actor and reason.	All document screens
ApprovalBar	Fixed bottom bar with the one to three actions the current role may take on this document in this state; disabled actions show why (permission, offline, state).	All document screens
ReasonCapture	Coded reason list in the shop language plus optional text; mandatory where the audit type demands it.	Void, refund, write-off, adjustment, credit-limit change

Component	What it is	Used by
AttachmentCapture	Camera capture with thumbnail strip; queued upload; shows “pending upload” honestly.	GRN discrepancy, claim, write-off
ScanField	Single focused input that accepts scanner, keyboard code, or weight; parses GS1 envelopes; shows the resolved item card with thumbnail.	Till sale, GRN, count
NumericKeypad	Large on-screen keypad with quantity/price/PIN modes; hardware keypad passthrough.	Till
ItemCard	Thumbnail, name in shop language (fallback marked), unit, price, MRP, batch/expiry badge, stock on hand.	Search results, picker, GRN, count
MoneyDisplay	Totals block: subtotal, reductions with rule name, total, tendered, change; 40 px total.	Till, customer display
FreshnessStamp	“Figures include shops synced up to hh:mm; 3 shops not yet synced” on every federation-level figure.	M8, exposure and dashboards
ExceptionQueue	List of REVIEW/ALERT events in scope with acknowledge action and disposition codes.	M1 (governance), M5, M6 supervisors
TrainingBadge	Unmistakable banner and colour shift when the shell is in sandbox mode (T-04).	Both shells

2.3 Usability rules for shop screens

- **Three taps to sell.** Scan, scan, tender. Nothing on the sales screen requires reading more than one word to act.
- **No modal chains.** A screen never opens a dialog that opens another; confirmations are inline in the ApprovalBar.
- **Destructive actions need a reason, never a second click.** A void asks why (coded), not “are you sure?”.
- **The keypad is the primary input.** Quantity, weight, PIN and price look-ups all work without a keyboard; search by name is the fallback.
- **Same place every time.** Basket left, item card right, totals bottom-right, actions bottom bar; the layout never reflows by mode.
- **Show what the system knows, honestly.** Offline state, pending uploads, stale ceilings and missing translations are visible, never hidden.
- **Training mode is loud.** Sandbox transactions look different enough that a real receipt can never be mistaken for practice (T-04).
- **Validate with real cashiers twice** (Stage 3 usability rounds) before any screen is frozen.

3. Web back-office shell

Aspect	Design
Stack	React 18+ with TypeScript; a single application with one route namespace per module (M1–M9); Vite build; component library implementing Section 2 on top of a headless primitives library (decision request DR-2 for the base library).
Authentication	OIDC authorization code flow with PKCE against the identity service (doc 19); access token in memory, refresh via the provider; step-up TOTP prompt rendered by the shell when an API call returns the step-up challenge.
Scope and navigation	The shell resolves the user's scopes and policy class once, renders the ScopeBanner and a navigation limited to modules the user's roles touch; a scope switcher only for users with more than one scope (Federation view, MPCS administrators of several shops).
Data access	A typed client generated from each module's OpenAPI slice (the contract file of doc 20 section 5); server-state caching with stale-while-revalidate; every list is server-paged and server-filtered; no client-side joins across modules.
Forms	Validation schema generated from the contract; inline errors in the shop language; drafts autosave; issued documents render read-only through the same components.
Permissions in the UI	Visibility from the resolved permission set; every action button also handles the server's permission denial gracefully (the server is the authority, doc 19 X-12).
Internationalisation	Message catalogue loaded per language; ICU MessageFormat; dates and money formatted centrally; language follows the user's location (doc 19 5.1).
Printing and export	A4 output is server-rendered PDF (doc 19 5.3) opened in a viewer; spreadsheet export is a server job with a download link.
Responsiveness	Document pages collapse to a single column; line tables become stacked cards on phones; approval bars stay fixed at the bottom.
Accessibility	Keyboard operable throughout; focus order follows the paper reading order; WCAG 2.2 AA targets.
Error and degraded states	Loss of connectivity shows a banner and keeps read views; write actions are disabled with the reason; a version banner appears when the deployed backend is newer than the loaded client (doc 31).
Module packaging	Each module contributes a route group, its screens, and its generated client; the shell owns layout, auth, i18n, navigation and the component library. A module developer adds screens without touching the shell.

4. Till application shell

4.1 Technology decision

Three stacks were weighed for the till: native Android (Kotlin, Jetpack Compose), Flutter, and React Native. The till needs device-owner and lock-task control, direct peripheral access, a durable local database with safe shutdown, raster printing, background sync and a two-day offline life on modest ARM hardware. Native Android is the only stack where all of these are first-class; the others reach them through bridges that become the project's risk.

Criterion	Native Android (Kotlin/Compose)	Flutter	React Native
Peripherals (scanner, printer, drawer, scale, display)	Direct SDK access	Platform channels; vendor SDKs still native	Native modules; same
Kiosk, device owner, silent update	First-class	Via native code	Via native code
Local database, transactions, fsync control	Room/SQLite with full control	sqflite/drift; adequate	Adequate; bridging cost
Sinhala/Tamil rendering and raster receipts	Platform text stack	Own text engine; shaping good	Platform text stack
Team fit for full-stack (React) developers	Learning curve	New language	Familiar
Long-term risk	Lowest	Moderate	Highest for hardware-heavy apps

Recommendation (DR-1): native Android for the till and driver app, built by a dedicated till track of two to three developers; module developers hand shop-mode screens to that track as specifications and reviewed Compose components, not as web code. This is the one place the “full-stack owns the module end to end” rule bends, and it bends deliberately.

4.2 Shell responsibilities

Aspect	Design
Device posture	Runs as the device-owner-managed kiosk app in lock-task mode; no launcher, no settings, no other apps (doc 31).
Local data	SQLite through Room; write-ahead logging; the same logical model as doc 18 for what the till writes; read-only cache tables for snapshots; per-transaction fsync for issuance (doc 18 5.5); database encrypted at rest with a key in the Android keystore.
Identity	PIN screen (doc 19 2.1); operator session; idle lock; device credential for sync.
Sync worker	Background job: upload outbox in sequence, download snapshots and apply atomically; exposes status to the ScopeBanner (last synced, pending count).
Peripheral layer	Interface per UnifiedPOS category (scanner, POS printer, cash drawer, scale, line display, label printer) with a simulator implementation for development and training; vendor SDK adapters behind it; health check per device on session open.
Receipt renderer	Composes the receipt as a bitmap using the platform text stack and Noto fonts; prints as raster; the same renderer produces Z-report, day-sales and till-session reports offline (O-02).
Scan layer	Symbology detection, GS1 element-string and Digital Link parsing, AI 01/10/17 extraction, dual-mark de-duplication (single beep), 20–29 prefix weigh-label decoding.
Modes	Sales mode and shop-management mode, role-gated, park-first, audited (doc 20 M6).
Training mode	Sandbox with the TrainingBadge, simulated peripherals, separate local database, no sync.
Telemetry	Version, last sync, crash reports, peripheral health reported on sync; feeds fleet dashboards (doc 31).

4.3 Sales screen layout (the one screen that matters most)

Zone	Content	Behaviour
Top strip	ScopeBanner: shop name, till number, operator, language, online/offline and pending-upload indicator, training badge	Always visible; never scrolls
Left (60%)	Basket: line list newest at the bottom, each line with thumbnail, name, qty × unit price, reductions with rule name, line total; a selected line exposes qty change and void-line	Scan focus always returns here; scrolling keeps the last line visible
Right (40%) upper	ScanField and ItemCard for the last scanned or searched item: thumbnail, name, price, MRP badge, stock; search results list when searching	Picker and negative-stock acknowledgement render here, not as dialogs
Right (40%) lower	MoneyDisplay: subtotal, reductions, total (40 px), then tender entry with keypad	Tender buttons: cash, card reference, account, voucher (later); change due shown large
Bottom bar	Park, Resume, Void, Refund (same session), Reprint, Supervisor, Mode	Disabled actions state why (offline, permission)

5. Screen inventories by module

The handover baseline. Surface: W = web, T = till (sales or shop mode), D = driver. Offline: whether the screen is usable without central. Each module design (docs 21–29) refines these into flows and wireframes; the inventory itself is frozen with the module's contract.

5.1 M1 – Party, Tenancy & Security

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Entity register	W	Federation admin	Society register	No	List and detail; relationships tab; onboarding status
Trading relationship	W	Seller admin/accounts	Agreement sheet	No	Terms, price list, credit limit (MFA), dispute window; effective-dated history
Locations and tills	W	Entity admin	Shop list	No	Primary till assignment (audited); device assignment; size band
Users	W	Entity/MPCS admin	Staff register	No	Create, deactivate, reset credential (MFA); scope selection
Role builder	W	Entity admin	Duties list	No	Clone template, tick permissions grouped by paper document, set limits; diff against template
External grants	W	Federation admin	Access letter	No	Time-boxed regulator/auditor grants; expiry
Exception queue	W	Named reviewers	Exceptions book	No	Shared component; disposition codes
Device enrolment	W	Entity admin	Asset register	No	QR provisioning token, assignment, suspend

5.2 M2 – Catalogue & Batch

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Catalogue browser	W	Steward, MPCS	Price book	No	Search in three scripts; status filter; thumbnails; tags
SKU editor	W	Steward, MPCS	Product card	No	Three names (EN mandatory), units and conversions, barcodes, images, tax, flags, attributes
Promote local SKU	W	Steward	–	No	Alias mapping preview showing history preserved
Bulk import and validation	W	Steward	Stock list	No	Upload, validation report, duplicate suspects across scripts, translation completeness
Batch register	W	Warehouse, MPCS	Batch card	No	Expiry, MRP, supplier; correction creates new batch
Control price register	W	Federation steward	Gazette	No	Effective-dated ceilings by SKU or tag; gazette reference; staleness
Item lookup	T	Cashier	Price list	Yes	Shared ScanField + ItemCard; read-only

5.3 M3 – Pricing & Rules

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Trade price list	W	Seller pricing	Wholesale price list	No	Per relationship; volume tiers; effective dates; ceiling validation errors inline
Retail price list	W	MPCS pricing	Shelf price list	No	One list per MPCS; label print job; advisory prices shown beside
Rule editor	W	MPCS pricing	Offer notice	No	Closed vocabulary as a form (kind, applies to, benefit, validity); preview against sample basket
Multi-MRP policy	W	MPCS pricing	–	No	Per SKU or tag: auto-lowest / barcode / picker
Markdown queue	W/T	Supervisor	Expiry list	T: yes	Batches entering the window; markdown rule stamp

Screen	Surface	Roles	Paper mirrored	Offline	Notes
MRP picker	T	Cashier	Pack label	Yes	Rendered in the ItemCard zone; shows batch, MRP, expiry and image side by side

5.4 M4 – Trading Documents

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Requisition (order) book	W	Buyer stores/manager	Requisition book	No	Suggested lines from reorder (P2) plus manual; seller availability; submit
Order desk (seller)	W	Seller sales	Order register	No	Accept with allocation and ETA; partial; backorders; lock window
Delivery note	W	Seller dispatch	Delivery note	No	Drops per vehicle; print; driver assignment
Drop list and proof of delivery	D	Driver	Signed DO	Yes	Stops in order, signature/photo, timestamp
Goods received (GRN)	T (shop mode) / W	Receiver	GRN book	Yes	Expected lines from DN; count actuals; auto-discrepancy; photos; confirm (ownership transfers)
Discrepancy and claim	W	Both parties	Claim form	No	Timeline, evidence, resolution, credit note issue
Invoice and credit/debit notes	W	Seller accounts	Tax invoice	No	Issued read-only; links to GRN/DN; print A4
Payment receipt and matching	W	Seller accounts	Receipt book	No	Select invoices to settle with amounts (I-07); MFA
Statement	W	Both	Statement of account	No	Per relationship, period; freshness stamp
Exposure view	W	Seller	Ledger summary	No	Warn-only; ageing buckets; freshness

5.5 M5 – Inventory, Costing, Repack & Loss

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Stock position	W/T	Stores, shop	Stock book	T: yes	By SKU and batch, expiry badges, on-hand and cost (cost hidden at till)
Movement ledger	W	Stores, auditor	Stock card	No	Per SKU: every movement with document link
Count task	T (shop mode) / W	Shop, stores	Stocktake sheet	Yes	Lines to count; enter actuals; variance shown after entry; submit
Adjustment approval	W	Approver	Adjustment voucher	No	Threshold escalation; MFA
Write-off request	T (shop mode) / W	Requester	Damage/expiry register	Draft: yes	Category, batch, qty, photos, witness selection
Write-off approval	W	Witness, approver	Same	No	Requester ≠ approver enforced visibly
Repack	W/T	Stores	Production sheet	T: yes	Recipe, input lot, actual yield; reversal while eligible
Intra-entity transfer	W	MPCS	Transfer note	No	Shop to shop at cost, MPCS-approved
Opening balance	W	Two parties	Opening stock sheet	No	Signed two-party document at onboarding

5.6 M6 – POS & Offline

Screen	Surface	Roles	Paper mirrored	Offline	Notes
PIN and session	T	Operator	Attendance book	Yes	Open with float; blind close; cash drop
Sales	T	Cashier	Till slip	Yes	Section 4.3
Park / resume	T	Cashier	–	Yes	Parked list with age
Refund (same session)	T	Cashier	Refund slip	Yes	Select from own session receipts; reason
Refund (cross session)	T	Supervisor	Refund slip	No	Online lookup by receipt number
Weigh and price	T	Cashier	Scale ticket	Yes	Scale read or manual weight; per-kg price
Reports	T	Supervisor	Z-report, day sheet	Yes	Print via raster renderer

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Shop-management mode home	T	Shop-in-charge	Shop diary	Yes	Tiles: receive goods, count, write-off, reprint, sync status
Sync and device status	T	Supervisor	–	Yes	Pending uploads, last snapshot version, peripheral health, app version
Training mode	T	All	–	Yes	Full sales flow on a sandbox database

5.7 M7 – Customers & Payment Recording

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Customer lookup	T/W	Cashier, accounts	Khata index	T: yes (cached)	By phone or id; account balance and limit
Customer register	W	Accounts	Customer card	No	Minimal PII; phone change with reuse check; consent note
Account and limit	W	Accounts (MFA)	Khata page	No	Limit, balance, ageing; statement
Charge to account	T	Cashier	Khata entry	Yes (capped)	Tender path; offline cap warning
Customer payment	W/T	Accounts, supervisor	Receipt	T: no	Settle specific charges
Banking record	W	Accounts	Bank slip	No	Deposit reference against takings

5.8 M8 – Reporting Pipeline

Screen	Surface	Roles	Paper mirrored	Offline	Notes
Dashboard	W	By scope	Day sheet	No	Fixed tiles with FreshnessStamp; drill to catalogue reports
Catalogue report viewer	W	By scope	Standard forms	No	Parameters, A4 render, spreadsheet export
Freshness and fleet	W	Federation	–	No	Which shops synced when; stuck devices
Analyst query	W	Federation analyst	–	No	Read-only SQL on the replica under RLS (doc 20 Appendix A)

5.9 M9 – Integration

Screen	Surface	Roles	Paper mirrored	Offline	Notes
API clients	W	Federation admin (MFA)	–	No	Register client, scope, rotate credential
Notification templates	W	Federation, MPCs	Notice text	No	Per event, three languages, preview
Journal export	W	Accounts	Journal	No	Period, entity, posting map preview, download

6. Handover pack for a module developer

What a full-stack developer receives when a module is handed over, so that nothing depends on hallway knowledge.

Item	Source	Status at handover
Module design document (docs 21–29)	Stage 2	At definition of done
Contract file: commands, queries, events, with the OpenAPI slice and generated client	Module design section 5	Frozen
Screen inventory with flows and wireframes	This document section 5, refined in the module design section 8	Frozen with the contract
Shell and component library	Platform pair	Released version pinned
Message catalogue ids for the module	Module design	English filled; Sinhala and Tamil in review
Test scenarios (doc 13 subset) and invariants to prove	Module design section 10	Named
Configuration register entries	Module design section 7	Listed with defaults
Definition of done	Doc 20 section 1, plus usability review for any shop screen	–

Boundaries at handover. A module developer owns the module's backend and its web screens end to end, and delivers shop-mode screen specifications and reviewed components to the till track. The platform pair owns both shells and the component library and accepts component change requests from modules on a weekly cadence.

7. Decision requests

#	Point	Proposed	Decider
DR-1	Till technology stack	Native Android (Kotlin, Compose) with a dedicated till track	Chamin, engineering lead
DR-2	Web component base library	A headless primitives library styled by the design system; no heavy opinionated kit	Platform pair
DR-3	Wireframing and prototype toolchain	Design tool with a component library mirroring Section 2; clickable prototype for the two cashier rounds	UX reviewer
DR-4	Customer display in phase 1	Yes where the terminal has a second screen; line, total, change only	Federation
DR-5	Driver app scope	Drop list, proof of delivery, photos; nothing else	M4 designer

End of document. Draft for design review; screen inventories are superseded by the module designs as each reaches definition of done.